Bow Valley College

Course Outline

DATA2300

Analyzing Data
Fall 2026 - Current

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Bow Valley College is situated in Treaty 7 territory. We honour the territories and Peoples of: the Blackfoot Confederacy, which includes the Siksika, Kainai, Piikani, and Amskapi Piikani (Blackfeet) First Nations; the Tsuut'ina First Nation; and the Îyâxe Nakoda, which includes the Goodstoney, Chiniki, and Bearspaw First Nations. We also acknowledge the historical Northwest Métis homeland and the Otipemisiwak Métis Government of Alberta, districts 4, 5, and 6 in Battle River territory.

DATA2300 Analyzing Data

COURSE DESCRIPTION

This course focuses on end-to-end analysis to describe and prepare data; conduct exploratory analysis; compare variables; test hypotheses; and generalize findings. You will apply and evaluate regression techniques (simple, multiple, probability models, and time-series), develop predictive solutions (framework selection, model building, regression trees, random forests/boosting, classification, and forecasting), and perform causal analyses (experiment design and analysis, regression with observational data, difference-in-differences, panel methods, and construction of appropriate control groups). You will document workflows for reproducibility, check assumptions and data quality, and translate technical results into clear recommendations for non-technical stakeholders. In a typical workplace scenario, you might clean and explore service-usage data, fit and diagnose multiple regression and classification models to forecast demand, and then assess the impact of a new scheduling policy using a controlled A/B test or a difference-in-differences approach on panel data. In this course, you will communicate uncertainty, limitations, and ethics (privacy, bias, responsible use), collaborate with peers and clients, and propose specific, measurable actions supported by evidence.

REQUISITES	Complete the following courses: • DATA1100 - Programming for Data Analysis (3) • MATH1900 - Mathematics for Data Analysis (3)
EQUIVALENTS	None
CREDITS	6
HOURS	90
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	No

COURSE COMPETENCY

COMPETENCY TITLE

Analyze Data

COMPETENCY STATEMENT

Data analytics professionals extract actionable insights from data by applying techniques such as descriptive statistics, regression analysis, predictive modeling, and causal inference. They interpret and communicate findings to stakeholders, while ensuring reproducibility of results.

COMPETENCY DESCRIPTION

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COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong students. In addition to the discipline-specific skills that students acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Conduct descriptive data analysis	2, 3, 5
2	Analyze Regression	2, 3, 5
3	Create a predictive model	2, 3, 5
4	Perform a causal analysis	2, 3, 5

LEARNING PATHWAY

**The time it takes learners to demonstrate competencies will vary. An example of a suggested schedule for learning and development is shown below. Learners will need to plan out their assessment attempts within their course. For additional information, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	LEARNING AND DEVELOPMENT PLAN
1	Module 1: Conduct descriptive data analysis
2	Module 1: Conduct descriptive data analysis
3	Module 2: Analyze Regression
4	Module 2: Analyze Regression
5	Module 2: Analyze Regression
6	Module 3: Create a predictive model
7	Module 3: Create a predictive model
8	Module 3: Create a predictive model
9	Reading Week
10	Module 4: Perform a causal analysis
11	Module 4: Perform a causal analysis
12	Module 4: Perform a causal analysis
13	Complete Competency Assessment
14	Complete Competency Assessment
15	Complete Mastery Assessment

ASSESSMENT

This course follows assessment on demand approach, in which students will be assessed, and receive structured feedback, and a personalized learning plan. Students will also receive differentiated support from an instructor based on their individual needs.

Students will have a variety of ways to demonstrate they have met the required competency through the demonstration of learning outcomes and criteria as laid out in the rubric. Students will have multiple (but not unlimited) attempts to prove competency. It is suggested that students plan out their assessment attempts within their course.

Students will have flexibility in how they satisfy course learning outcomes while still adhering to the criteria found in the rubric and the Course Offering information. Please refer to the Course Offering Information and the rubric in Brightspace for additional information.

COURSE LEARNING OUTCOMES	ASSESSMENT
1-4	Competency Assessment (4 parts)
1-4	Mastery Assessment (optional)

PERFORMANCE STANDARDS

Successful demonstration of all learning outcomes is required to pass this course. This course may be part of a competency or larger program of study.

Please consult with the program area or the Bow Valley College website for completion requirements.

GRADING SCHEME

Symbol	**Description**	**Grade Point Value**
MAST	Mastery	4.00
COMP	Competent	3.00
DEVL	Developing	N/A
F	Fail	0.00
W	Withdraw	N/A

REQUIRED LEARNING RESOURCES

Békés, G., & Kézdi, G. (2021). Data analysis for business, economics, and policy. Cambridge University Press.

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Students with documented disabilities (e.g., mental health diagnosis, learning disability, physical disability, etc.) may qualify for classroom and/or exam accommodations. The College can also assist students who may be experiencing difficulties with learning but do not have a formal diagnosis. Students who need academic and exam accommodations based on other protected grounds may also make those requests.
It is the student's responsibility to request academic accommodations. Students are encouraged to contact Accessibility Services as early as possible.

Visit: Learner Success Services- 1st floor south campus
Contact: accessibility@bowvalleycollege.ca or 403-410-1440
BVC Central: Accessibility Services

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Students are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Student Code of Conduct (500-1-1), Student Appeals (500-1-12), Attendance (500-1- 10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13).
Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by students is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Academic Integrity:

At Bow Valley College, we value academic integrity. This means everyone at the college, including students, faculty, and staff, follow principles that foster ethical and truthful ways of being in their work and studies. The college is committed to integrity being infused in all aspects of work and studies. It is important to act with integrity in everything you do at the college. The values you practice at the college are the same values that will inform and guide your career. If someone breaks from the principles that are in agreement with academic integrity policy, the college is committed to carefully investigating and fairly handling any issues related to academic integrity and academic misconduct.

Academic misconduct is acting in such a way as to give yourself or others an unfair academic advantage or disadvantage. Examples of this can include:

- Plagiarism
- Cheating on tests or exams
- Tampering with or falsifying any grades or documents
- Sharing assignments or exam information with others
- Aiding others in dishonest acts
- Using artificial intelligence in graded work without the explicit consent of the instructor
- Recording anything in the classroom or other instructional area without permission

Breaking from the principles that guide academic integrity can lead to educative interventions and/or punitive measures. It is the student's responsibility to review the Academic Integrity Policy. Please review the following resources:

- Academic Integrity Policy Summary
- Academic Integrity Policy
- Academic Integrity Procedure

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student work. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where either the course instructor, a representative of Bow Valley College, or a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;

b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, https://www.bowvalleycollege.ca/academic-calendar.
Students are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements. You may use the recordings only for educational purposes. You must not copy, share, or use the recordings for any other purpose without the instructor's express permission. Your personal information is collected in accordance with section 4 (c) of the Protection of Privacy Act, Alberta (POPA) to deliver academic programming, support student flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. Bow Valley College uses automated systems to generate content and make decisions, recommendations and predictions, and your personal information may be included in these automated systems.
Your personal information will be used and disclosed in accordance with sections 12 and 13 of POPA and protected in accordance with section 10 of POPA. If you have any questions about the collection, use, disclosure or protection of personal information, please contact the Access and Privacy Officer at privacy@bowvalleycollege.ca.